

Isaac Kobby Anni

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Professional Summary

Ph.D. candidate in Data Science with research and applied experience spanning machine learning, natural language processing, predictive analytics, and AI system development. Current research centers on **SLM-First Agentic Systems**, with emphasis on privacy-preserving, cost-efficient architectures built on small language models for automated data science workflows. Experience includes academic research, student success analytics, NLP research collaboration, computer vision applications, and industry data science work across finance and geospatial settings. Strong record of building practical data pipelines, predictive models, analytical dashboards, and research-driven AI solutions.

Education

Bowling Green State University Ph.D. in Data Science	<i>Bowling Green, Ohio</i> Expected 2027
Bowling Green State University M.S. in Data Science	<i>Bowling Green, Ohio</i> May 2024
African Institute for Mathematical Sciences (AIMS) M.Sc. in Big Data, Minor in Statistics	<i>Dakar, Senegal</i> July 2022
University for Development Studies B.Sc. in Finance, Minor in Mathematics	<i>Navrongo, Ghana</i> June 2019

Research Interests

Small Language Models (SLMs), Agentic AI, Automated Feature Engineering, Case-Based Reasoning, Multi-Agent Systems, Natural Language Processing, Efficient AI, Privacy-Preserving AI, Predictive Modeling, Applied Machine Learning

Technical Skills

Programming: Python, R, SQL, C++, Java

Machine Learning / AI: Natural Language Processing, Transformer Models, Predictive Modeling, Feature Engineering, Computer Vision, Multimodal Learning, Classification, Clustering, Time Series Analysis

Frameworks / Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face, pandas, NumPy, OpenCV, Streamlit

Analytics / BI / Data Tools: Power BI, Tableau, Jupyter, Git, Docker, MongoDB, SQL/NoSQL systems

Methods: Statistical Modeling, Experimental Design, Causal Inference, Regression, Evaluation and Validation, Data Pipeline Development

Professional Experience

Graduate Assistant / Data Scientist, Student Success Analytics & Technologies Bowling Green State University	<i>Aug. 2022 – Present</i> Bowling Green, Ohio
- Develop predictive and analytical solutions using Python, SQL, and Power BI to support student success, retention analysis, and institutional decision-making. - Build and maintain data pipelines, dashboards, and reporting workflows for university stakeholders, translating large educational datasets into actionable insights. - Contribute to applied machine learning and statistical analysis projects involving student outcomes, persistence, and operational analytics. - Support production-oriented analytics workflows, including automation, dashboard standardization, and secure reporting practices.	
Research Fellow, NLP / Research Collaboration Computational Language Understanding / ICRISAT (remote collaboration)	<i>Mar. 2022 – Aug. 2022</i> Tucson, Arizona / Dakar, Senegal

- Contributed to research in natural language understanding and information extraction in collaboration with the Computational Language Understanding Lab and research partners.
- Worked on text-focused research pipelines, annotation-related tasks, and literature-driven NLP efforts contributing to collaborative publication work.
- Participated in research software and experimental workflows supporting reproducible NLP investigation.

Data Scientist Jr.

Oct. 2021 – Mar. 2022

Baamtu Technologies

Dakar, Senegal

- Built deep learning workflows for image-related applications, including segmentation and computer vision tasks on satellite and geospatial data.
- Integrated model outputs into interactive analytical tools and dashboards to support applied use cases and user-facing decision support.
- Worked on data preprocessing, training pipelines, and deployment-oriented experimentation for machine learning solutions.

Data Scientist Jr.

Jul. 2021 – Mar. 2022

Obertys Technologies

Dakar, Senegal

- Supported credit scoring and financial analytics projects through machine learning model development, feature engineering, and data analysis.
- Worked across multiple datasets to improve risk modeling workflows and support more data-informed lending decisions.
- Contributed to model interpretation, experimentation, and reporting for business and technical stakeholders.

Research and Selected Projects

Dissertation Research: SLM-First Agentic Systems for Automated Data Science

Researching small-language-model-first architectures for automated data science, emphasizing privacy-preserving, cost-efficient, and deployable AI systems. Current directions include automated feature engineering, agentic case-based reasoning, and orchestrated multi-agent workflows for end-to-end data science tasks.

Student Success Predictive Analytics

Applied machine learning, analytics engineering, and dashboard development to higher education data for institutional reporting and outcome-focused decision support.

NLP and Text Representation Research

Worked on text classification and document representation research, including comparative study of representation methods for downstream classification tasks.

Computer Vision and Applied AI Solutions

Contributed to practical machine learning applications involving image analysis, segmentation, and analytics integration in industry settings.

Publications

- **Anni, Isaac Kobby**, and Venu G. Dasigi. 2025. “Analysis of Document Representation for Text Classification.” In *Advances in Information and Communication*, Future of Information and Communication Conference (FICC 2025), pp. 261–273. Springer Nature Switzerland.
- Alexeeva, Maria, Caroline Hyland, Keith Alcock, Allegra A. Beal Cohen, Hubert Kanyamahanga, **Isaac Kobby Anni**, and Mihai Surdeanu. 2023. “Annotating and Training for Population Subjective Views.” In *Proceedings of the 13th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis*, pp. 416–430. Association for Computational Linguistics.

Manuscripts and Ongoing Research

- SLM-First Agentic Systems for Automated Data Science (dissertation-centered research agenda).
- Ongoing work on efficient, privacy-aware, and deployable small language model systems for automated feature engineering and multi-agent AI workflows.

Academic Service

- Peer Reviewer, *The 20th International AAAI Conference on Web and Social Media*.
- Peer Reviewer, *The Second Bangla Language Processing (BLP) Workshop*

Awards, Presentations, and Professional Activities

- Ph.D. research focused on efficient and privacy-preserving AI systems using small language models.
- Research collaboration experience spanning Bowling Green State University, University of Arizona collaboration contexts, and international research settings.
- Open to adding invited talks, poster presentations, workshops, fellowships, or awards here if you want a stronger NIW-facing version.

Professional Profiles

- Personal Website: isaackobby.com
- LinkedIn: linkedin.com/in/isaac-kobby-anni
- Google Scholar: scholar.google.com/isaac-kobby-anni